Workshop on Research Data Management and data protection in the Social Sciences

Please note that the workshop will be recorded.

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Welcome and housekeeping rules

- Please mute your microphone when you are not speaking
- Please turn off your camera during presentations - the session is being recorded
- Post your questions in the chat or raise your hand using the "raise your hand" function in Zoom ("Reactions" button)
- Need help? Write in the chat!
- Tell us how we did in the survey (more about that later)
- The materials (including the presentations) will be shared afterwards

AUSSDA…

- AUSSDA – the Austrian Social Science Data Archive
- Consortium of the universities of Vienna, Graz, Linz and Innsbruck
- AUSSDA's mission is to make social science data FAIR – findable, accessible, interoperable and reusable
- AUSSDA archives research data (such as representative population surveys) and makes them available in common data formats (such as SPSS, Stata, R) for subsequent reuse
- Services open to all researchers

CESSDA

- AUSSDA as Service Provider in CESSDA ERIC - the Consortium of European Social Science Data Archives (CESSDA ERIC).
- A European Research Infrastructure Consortium (ERIC)
- brings together social science data archives across Europe
- for a joint approach to research data management in the social sciences

Who we are

- Anita Bodlos: data curation
- Veronika Heider: data curation
- Lisa Hönegger: RDM consultations, data protection & licences

Agenda

- 10:00 – Welcome & Intro about AUSSDA
- 10:05 – Data management plans (DMP) in the Social Sciences
- 10:45 – Data protection in the Social Sciences

11:20 Coffee break

- 11:30 – Pseudonymization strategies used by AUSSDA
  STATA code for pseudonymizing your data
- 12:30 – End

Data management plans in the Social Sciences

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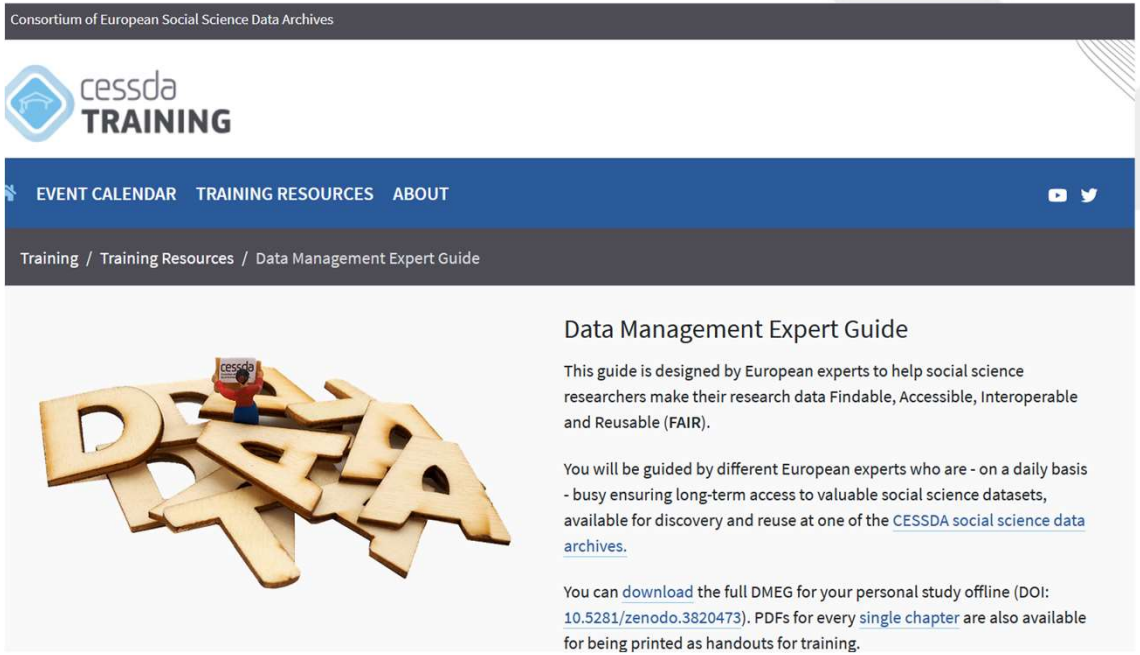
Data Management Expert Guide (DMEG)

- A comprehensive guide for data management in the Social Sciences
- Website: https://www.cessda.eu/Training/Training-Resources/Library/Data-Management-Expert-Guide
- Document: CESSDA Training Team (2017 - 2019). CESSDA Data Management Expert Guide. Bergen, Norway: CESSDA ERIC. DOI: [10.5281/zenodo.3820473](https://doi.org/10.5281/zenodo.3820473)



What we will talk about in this part

- Research data management (RDM) and FAIR data
- Data management plans (DMPs)
  - Their purpose
  - Their content (mainly following the FAIR criteria)

What is RDM?

„Research data management refers to how you handle, organise, and structure your research data throughout the research process.”

(DMEG, p. 9)

FAIR data

Findable → assigning PIDs, enriching with meta data, using standardized vocabularies, registering data in a searchable catalogue, etc.

Accessible → transparent access options, AAI, as open as possible etc.

Interoperable → allow interoperability with other data, tools and systems through API, proper documentation or different and common formats, etc.

Reusable → clear reuse regulations (licensing), additional documentation materials, etc.

Repositories cover most aspects

Hönegger 2021: Research Data Management in the Social Sciences

Data management plans (DMPs)…

- … contain all relevant aspects of RDM
- … are a formal document
- … are a „living document"
- … are a mandatory deliverable for many grants
- … can vary in their content depending on your research data and requirements by funders

DMEG: p. 9-10

What are benefits of a DMP?

- Plan for the future
  - How do you organise RDM?
  - Which problems will you likely encounter and how can you get ahead of them?
  - Good RDM facilitates the documentation, publication and reuse of data

DMEG: p. 10-12

What are benefits of a DMP?

- Guidance throughout the project
  - One document covering all aspects of RDM
  - Guidelines on directories, filenaming, versioning, backup-strategies
  - Administrative information:
    - Grant numbers, team members…
    - References to further documentation material (research logs, method reports…)

DMEG: p. 10-12

What are benefits of a DMP?

- Plan your budget
  - Estimate the budget for RMD, resources in terms of money and staff, storing data….

DMEG: p. 10-12

What are benefits of a DMP?

- Consider the FAIRness of your data at the beginning of your project
  - Which repository to use for publications
  - Data formats
  - Licences
  - ….

DMEG: p. 10-12

What are benefits of a DMP?

- Demontrate your commitment to RDM and sustainable data handling

DMEG: p. 10-12

Templates and collections

- Templates/checklists for DMPs
  - CESSDA: https://www.cessda.eu/Training/Training-Resources/Library/Data-Management-Expert-Guide, https://www.cessda.eu/content/download/4302/48656/file/TTT_DO_DMPExpertGuide_v1.2.pdf
  - H2020: https://ec.europa.eu/research/participants/data/ref/h2020/grants_manual/hi/oa_pilot/h2020-hi-oa-data-mgt_en.pdf
  - Horizon Europe: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/programme-guide_horizon_en.pdf, p.41
  - FWF: https://www.fwf.ac.at/fileadmin/files/Dokumente/Open_Access/FWF_DMPTemplate_d.pdf
  - Science Europe: https://scienceeurope.org/our-resources/practical-guide-to-the-international-alignment-of-research-data-management/
- Collections of DMPs:
  - DCC: https://www.dcc.ac.uk/resources/data-management-plans/guidance-examples
  - Phaidra: https://phaidra.univie.ac.at/detail/o:1140797
- Data Horror Escape Room: https://sites.google.com/vu.nl/datahorror

Example 1: H2020 template

https://ec.europa.eu/research/participants/data/ref/h2020/grants_manual/hi/oa_pilot/h2020-hi-oa-data-mgt_en.pdf

6 chapters:

1. *Data Summary*
2. *FAIR data*
3. *Allocation of resources*
4. *Data security*
5. *Ethical aspects*
6. *Other issues*

We will use this structure to discuss the content of a DMP

Example 2: CESSDA Data Management Expert Guide

https://www.cessda.eu/Training/Training-Resources/Library/Data-Management-Expert-Guide

Data Management Expert Guide

1. Plan
2. Organise & Document
3. Process
4. Store
5. Protect
6. Archive & Publish
7. Discover
8. Contributors

Content of DMPs

Some administrative information to include

- Author(s) of the DMP
- Who is reponsible for updating the DMP?
- Date and version
- Description of the project
- Start and end date
- PIs, contact details of researchers and important collaborators
- Information about funding and grants
- Who „owns" the data?
- …

DMEG: p. 29-30

1. Description of your research data

Description of your research data

Questions of the H2020 template (p.7):

- *What is the purpose of the data collection/generation and its relation to the objectives of the project?*
- *What types and formats of data will the project generate/collect?*
- *Will you re-use any existing data and how?*
- *What is the origin of the data?*
- *What is the expected size of the data?*
- *To whom might it be useful ('data utility')?*

HE template:

- Additionally: more emphasis on whether you re-use data

Description of your research data

Ways to describe your research data:

- "Type of data"
  - e.g. DDI Controlled Vocabulary for Kind of Data Format: https://ddialliance.org/Specification/DDI-CV/KindOfDataFormat_1.0.html
- Data formats
  - Proprietary and non-proprietary formats
- "Size & Complexity"
- "Research phase"
- (Sensitive) personal data
- Quantitative/qualitative data

DMEG: p. 13-18

2.1 Findability of data

Findability of data

"Making data findable, including provisions for metadata" (H2020 template: p.7) regarding:

- persistent identifiers (PIDs)
- "(N)aming conventions"
- Keywords
- Version control
- Metadata and metadata standards

HE template:

- More emphasis on the interoperability of metadata

Take advantage of repositories!

Findability: Repositories

How to choose?

1. „A (trusted) domain repository" (e.g. AUSSDA)
2. Your institution's repository
3. Catch-all, free repository (e.g. Zenodo)
4. Use databases for finding a repository:
    1. Re3data.org
    2. https://forschungsinfrastruktur.bmbwf.gv.at/de

DMEG: p. 162; OpenAIRE: https://www.openaire.eu/opendatapilot-repository-guide

Persistent identifiers (PIDs)

Usually provided by repositories

- Digital Object Identifiers (DOI)
- URN
- Handle

Naming conventions

- Data files
- Variable names and labels
  - Esp. for connected data files

DMEG: ch.2

Version control

- Use version (v1_0, v1_1) and/or the date in the filename
- Documentation about the versions
  - E.g. in a version control table or research log
  - Who edited which version, when and why?
- Repositories may manage version control for your research data

DMEG: p. 80-81

Rich metadata

Often: provided by depositors, managed/supportend by repositories

- Meaningful abstract
- Keywords, topics, kind of data, data sources, collection mode, time method, unit of analysis…..

Example from AUSSDA: https://doi.org/10.11587/DEHBL2

Metadata standards

- Ask your repository

Findability: AUSSDA…

- …is certified with the Core Trust Seal.
- …is harvested by the CESSDA data catalogue.
- …provides DOIs for your data.
  - can be reserved as soon as the contracts are signed.
- …takes care of naming conventions of files.
- …checks your metadata and gives feedback.
- …supports you during updates and management of version control.
  - Dataverse documents all changes.
- …uses DDI and CESSDA metadata standards.
  - For more information on metadata see the self-deposit manual: https://aussda.at/fileadmin/user_upload/p_aussda/Documents/Self-Deposit_Manual_v1_0.pdf

2.1 Accessibility of data

Which data will you (not) share?

- Reasons for not sharing data:
  - Data protection
  - Cooperations with third parties
  - Intellectual property rights
- Discuss your reasons in your DMP

H2020 template: p.7; DMEG: ch.5

Repositories

HE template, p. 4:

- *Will the data be deposited in a trusted repository?*
- *Have you explored appropriate arrangements with the identified repository where your data will be deposited?*
- *Does the repository ensure that the data is assigned an identifier? Will the repository resolve the identifier to a digital object?*

Software

- "What methods or software tools are needed to access the data?"
- If possible: publish the code for accessing the data along with the data (e.g. syntax for reading in csv-files to STATA/R)

H2020 template: p.7

Access conditions

- Depend on your kind of research data:
  - Data protection often requires access restrictions
  - Data do not have to be „Open Access" to be FAIR!

… more information in the next part of the workshop on data protection and legal aspects

Metadata

Emphasised in HE template, p.4-5:

- Published as CC0?
- Contain access conditions?
- Long-term available?
- Contain information about software?

2.3 Interoperability of data

H2020 template: p.8, DMEG, ch.8

Data formats

- Which data formats do you use for publishing your data and why?
  - E.g. SPSS (.sav) or STATA (.dta) because they are common in Social Sciences
  - E.g. .R or .csv because they are open source/non-proprietary

- AUSSDA publishes all data as .zsav, .dta and .csv

H2020 template: p.8, DMEG p.84

Interoperability of data

- Do you use any standardized codes in your data?
  - E.g. ISCED, ISCO, NUTS codes
- Can your research data be connected to other research data?
  - E.g. coding schemes of other research (e.g. CMP or AUTNES codes)

H2020 template: p.8, DMEG ch.3.2, p.84

Standard/controlled vocabularies

- Check your repository: which controlled vocabularies do they use?
- Controlled vocabulary used by AUSSDA:
  - ELSST - The European Language Social Science Thesaurus https://elsst.cessda.eu/
  - CESSDA Topic Classification https://vocabularies.cessda.eu/vocabulary/TopicClassification
  - DDI Data Documentation Initiative https://ddialliance.org/

H2020 template: p.8, DMEG p.84

2.4 Reusability of data

H2020 template: p.8

Licences

- Will be covered in the next part of the workshop on data protection and legal aspects

Time of publication

- When do you plan to publish your data?
  - Reach out to your repository asap
- Do you plan to use an embargo? Why?
  - Check funding regulations
  - Check with your repository if this service is provided

H2020 template: p.8

Availability of your data

- Do you plan a time limit to the availability of your data? Why?
- Which measures do you take to make your data reusable for a long time period? e.g.:
  - Publication in a repository
  - Publication in a format frequently used for long-term preservation (e.g. .csv, PDF/A)

H2020 template: p.8

Quality of data

Describe your measures to increase data quality, e.g.

- Data gathering:
  - coding rules, coder training, reliability tests, data entry checks, check of transcripts…
- Sanity checks of data:
  - Plausibility of value range and observations

H2020 template: p.8, DMEG: ch.3

Documentation material

- Emphasised in the HE template (p.5)
- The more documentation, the higher the reuse potential
- Kind of documentation material: details in the third part of the workshop on AUSSDA deposit guidelines

3. Allocation of resources

H2020 template: p.8

FAIRness of data has a price

- Who will be in charge of RDM/making your research data FAIR?
- How much resources do you allocate for RDM?
- Repositories support depositors in many ways, but depositors must
  - prepare the data
  - prepare the documentation material
  - handle all administrative and legal questions
  - be available during the data curation process
  - incorporate the changes recommended by the repositores

H2020 template: p.8

Costs of data publication

- Many repositories publish data free of charge
- If you host your own data:
  - Who is in charge of the data after the end of the project?
  - Did you allocate the necessary resources for long term preservation, incl. storage costs, handling access requests, conducting updates etc.?

H2020 template: p.8

4. Data security

H2020 template: p.8

Data security

- Backup strategies, e.g.
  - you use your organisation's IT infrastructure and rely on your organisation's backup strategies
- Secure storage, e.g.
  - data are accessbile only via computers at your organisations
  - all devices are password protected (e.g. by a password policy)
  - you use encrypted devices and folders (e.g. Bitlocker, Veracrypt)

H2020 template: p.8

Data security

- Exchanging data
  - E.g. you use secure filesender like the Aconet filesender
- After publication
  - E.g. you rely on your repository's security procedures and sustainability

H2020 template: p.8

5. Ethical aspects

Ethical aspects

- "Are there any ethical or legal issues that can have an impact on data sharing?" e.g. personal data, consent forms

Will be covered in the next part on data protection and legal aspects

H2020 template: p.9

„Other research outputs“

HE template, p. 5:

- *In addition to the management of data, beneficiaries should also consider and plan for the management of other research outputs that may be generated or re-used throughout their projects. Such outputs can be either digital (e.g. software, workflows, protocols, models, etc.) or physical (e.g. new materials, antibodies, reagents, samples, etc.).*
- *Beneficiaries should consider which of the questions pertaining to FAIR data above, can apply to the management of other research outputs, and should strive to provide sufficient detail on how their research outputs will be managed and shared, or made available for re-use, in line with the FAIR principles.*

Q & A

Data protection in the social sciences

Legal basics, challenges and suggestions to manage your data

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What we will talk about today

- Legal basics & their challenges
- Open Data vs. data protection
- Making data sharing possible
- Data protection in the DMP
- Suggestions to manage your data

Legal basics & their challenges

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What is personal data?

- Social science data is often personal data -> GDPR applies

- Personal data:

  'personal data' means any information relating to an identified or identifiable natural person ('data subject'); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person; (Art. 4 (1) GDPR)

- Pseudonymized data = personal data
- Anonymous data ≠ personal data

Classification of (personal) data

- Classification in personal data (non-pseudonymized), pseudonymized data and anonymous data is a challenge in the social sciences
  - personal data (non-pseudonymized): data with direct indentifiers
  - pseudonymized data: data without direct and indirect identifiers – but with key (=additional information)
  - anonymous data: data without direct and indirect identifiers – no de-identification possible

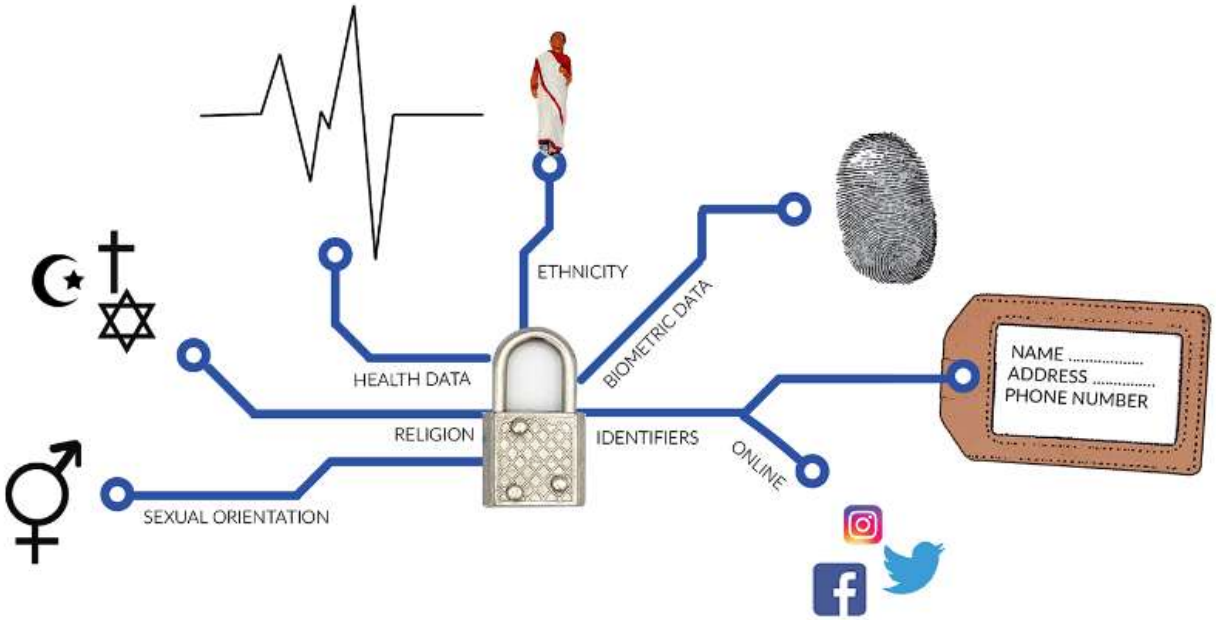
Classification of (personal) data

- Indirect identification by sheer volume of data or special characteristics of target groups

- Difficult question: When is anoymization (pseudonymization) reached?
  - When individuals cannot be de-identified (without key)
    - To consider: „all the means reasonably likely to be used ... to identify the natural person, account should be taken of all objective factors, such as the costs of and the amount of time required for identification" (GDPR)

Sensitivity of personal data

- The GDPR protects all personal data
- Special measures for „sensitive" data:
  - Special categories of personal data: necessary to provide for suitable and specific measures to safeguard the fundamental rights and the interests of the data subject. (Art. 9 (2) lit. j. GDPR)



Source: https://www.cessda.eu/Research-Infrastructure/Training/Expert-tour-guide-on-Data-Management

Data protection in the EU and Austria

- GDPR
- DSG (Data protection Act), FOG (Research and Organisation Act), other laws
- Institutional guidelines, policies, codes of conduct
  - Often helpful information about legal or ethical aspects of data management (i.e. regulating usage rights of data (IPR) or the right to license the data)
  - Or information on useful services, consultations, templates available etc.

→ One European framework but still national and institutional regulations – consider this in research cooperation projects

- European diversity in data protection: https://www.cessda.eu/Training/Training-Resources/Library/Data-Management-Expert-Guide/5.-Protect/Processing-personal-data/Diversity-in-data-protection

Open data vs. data protection

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Open data vs. data protection

- Discrepancy between regulations on open data and data protection

- The legal side of it:
  - Funding regulations – and how to deal with them
- Ethical aspects to consider:
  - Good scientific practice, preservation of knowledge and transparency

Funding regulations - EU

- H2020 early stage: Open research data (ORD)/DMP Pilot
- H2020 later on: ORD/DMP by default (exceptions)
- **Horizon Europe**:
  - From open to FAIR data and RDM
  - No opting out of RDM – all data can be managed
  - According to FAIR principles – no discrepancy to data protection
    - "as open as possible as closed as necessary"

  → **From open by default to FAIR**

Funding regulations – FWF

- "Open access is mandatory for research data on which the research publications of the project are based … These data should be published as soon as possible, but at the latest together with the corresponding research publication. If, for legal, ethical or other reasons, open access to these data is not or only partially possible, this must be explained in the Data Management Plan (DMP)"

https://www.fwf.ac.at/en/research-funding/open-access-policy/open-access-to-research-data

Conclusion: similar as in Horizon Europe – RDM is an obligation, not open data

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Making data sharing possible

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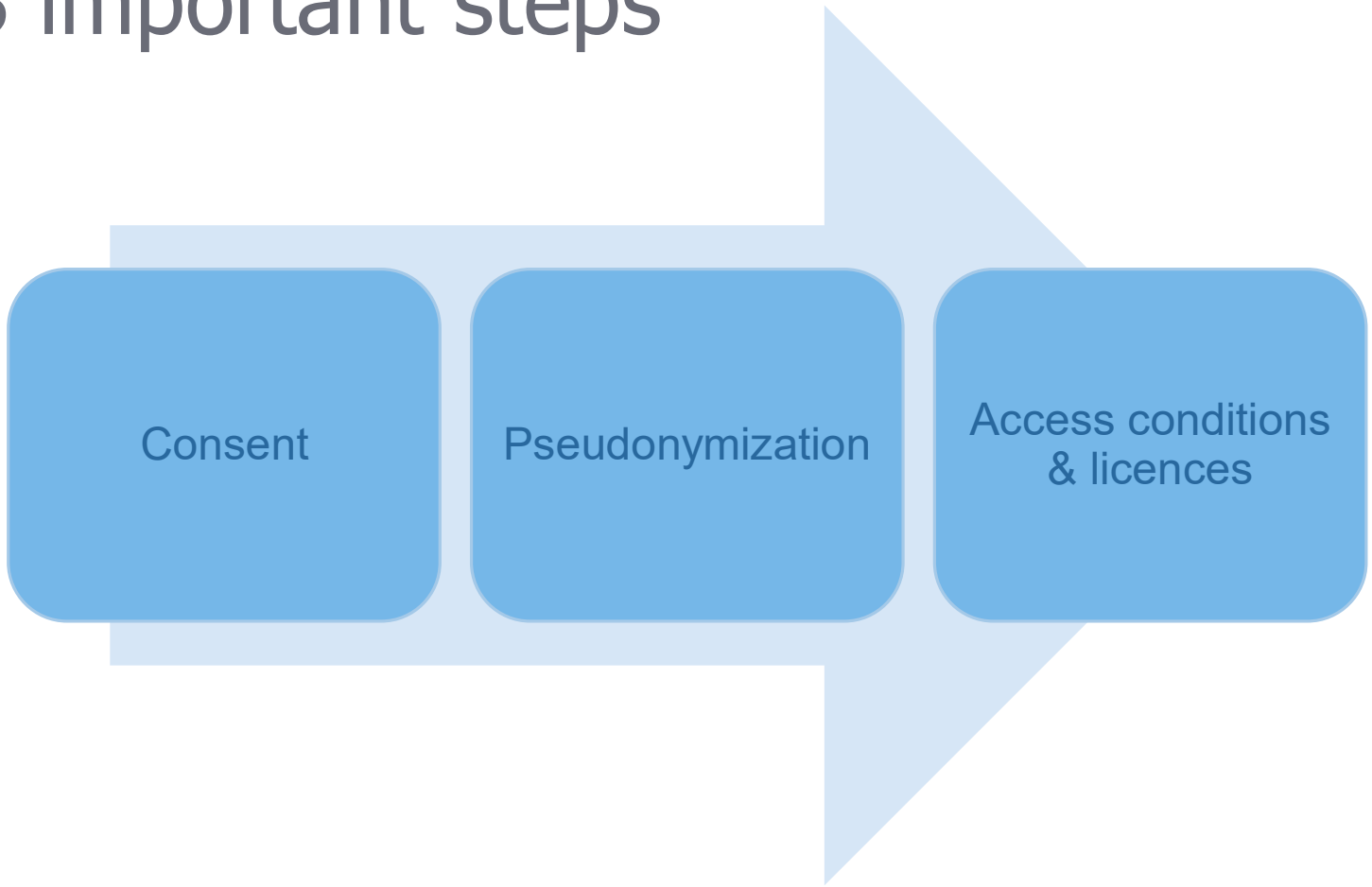
How to make data sharing possible – 3 important steps



Informed Consent

- Legal basis to process personal data (Art. 6 lit.a GDPR)
  - giving information: which data is collected and how is it processed?
  - getting active and voluntary consent: based on comprehensive and easy to understand information
  - **To include: information about the archiving and publishing of data**

- Ethical aspect of research
  - Informing participants on how their data is used and what rights they have (even if consent is not the legal basis of processing)

Informed consent for data sharing - examples

"The data collected will be used exclusively in the research project and will not be passed on to third parties. After completion of the project, the collected data will be deleted."

"The data and information collected by this project will not be published in any individualized form and individual details from the study will not be disseminated in any form in order to protect your privacy and the privacy of other study participants. Only aggregated summary results without identifying information will be communicated."

"The data will only be published in aggregated form."

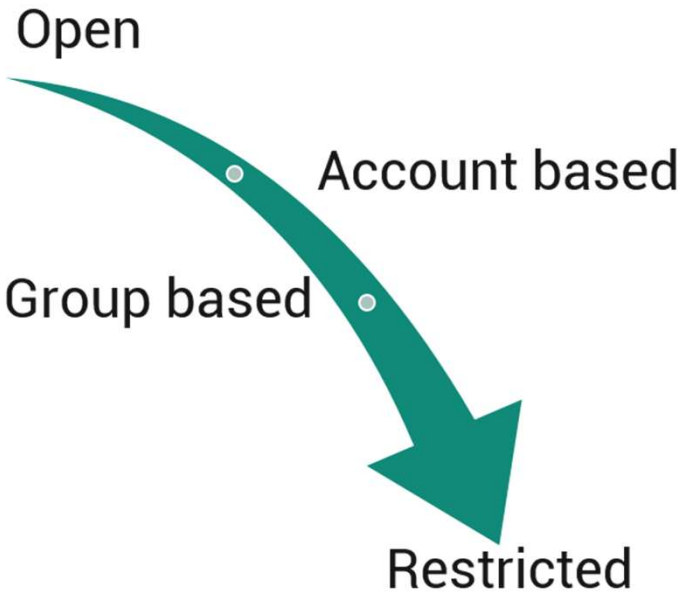
"To protect your privacy the data will be pseudonymised. That means that it is not possible for people external to the research project to identify you. Access to the non-pseudonymised data will be limited to the researchers involved in this project. In addition, the pseudonymised data will be transferred to AUSSDA, a social science data archive, who will archive the data and make it available for scientific reuse."

Pseudonymization of personal data

- Removing direct and indirect identifiers by:
  - Deleting the information
  - Substituting the information with a pseudonym (more relevant for qualitative data, e.g. interview transcripts)
  - Aggregating the information (more relevant for quantitative data, e.g. numeric data set)
- Store the key for de-identification safe and separate
  - Ensuring adequate technical and organisational measures (TOMs)

Access restrictions – making data available for scientific reuse

- Restricting access to enable data sharing
- Following the principle: „as open as possible – as closed as necessary"
- Raising awareness of allowed use of data and data protection



Licences

- Important to clarify the usage rights - otherwise reuse of data is not possible
- Licensing can easily be done by the rights holder
- Data: Licensing (sensitive) personal data for Scientific Use
  - With Scientific Use licences
    - i.e. AUSSDA SUF licence: https://aussda.at/fileadmin/user_upload/p_aussda/Documents/AUSSDA_License_Scientific_Use_EN.pdf

- Documentation: usually does not contain (sensitive) personal data
  - Should be licensed openly (i.e. with a CC BY licence)
- Metadata: should be licenced as open as possible – for optimal reuse license CC 0 (or public domain dedication)

Data protection in the DMP

Q&A

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Q&A from H2020, Horizon Europe and FWF templates

- **Can I write in the DMP that I will archive the data at AUSSDA?**
  - Theoretically yes, but please check with us first, if your data is suitable for archiving and fits our criteria before submitting a DMP. Also, don't be too specific about details that can only be decided after data collection or after data checks, i.e. the specific licence or the access condition to your data.

  - When writing the DMP and planing your data management - get in touch with us!

Q&A from H2020, Horizon Europe and FWF templates

- **Is the data safely stored in certified repositories for long term preservation and curation?**
  - It should be. If possible, use a certified and disciplinary repository (i.e. CTS certified) so the needs of your data can be met. If you deposit with AUSSDA, the data will be curated according to international and discipline specific standards and regulations and will be preserved long term.

Q&A from H2020, Horizon Europe and FWF templates

- **Who owns the data?**
  - It depends. But technically, ownership of data is highly disputed. Find out who has the usage rights to your data for licensing, publishing, reusing. And keep in mind that (especially in larger projects) it can take time to sort that out.

Q&A from H2020, Horizon Europe and FWF templates

- **Are there any legal barriers to making the research data fully or partially accessible?**
  - Probably yes – since personal data needs to be protected. However, with the appropriate measures, data can be made FAIR. We make data available "as open as possible, as closed as necessary".
  - Also think about other legal barriers, not just data protection - IPR (for example if you include other text or fotos); terms of use of sources you used (i.e. from social media), licences for data that you reused

Q&A from H2020, Horizon Europe and FWF templates

- **Are there any restrictions on the re-use of the data? If certain datasets cannot be shared (or need to be shared under restricted access conditions), explain why.**
  - There can be various forms of re-use restrictions of the data, most commonly the data can only be used for scientific purposes if (sensitive) personal data is included, because of data protection regulations.

Q&A from H2020, Horizon Europe and FWF templates

- **Will all data be made openly available?**
  - It depends on the data – social science data can often not be made available openly due to data protection regulations. The data should be made available as openly as possible. Documentation and metadata usually can be made available openly.

Q&A from H2020, Horizon Europe and FWF templates

- **Which licences will be applied to my data?**
  - There are different options. The selection of licence depends on various factors, among else the content and curation level of your data, as well as the source of your data. Please refrain from writing that all data will be made available open access via AUSSDA, as this is rarely the case with social science data.

Q&A from H2020, Horizon Europe and FWF templates

- **Will metadata be made openly available and licenced under a public domain dedication CC0? If not, please clarify why.**
  - Yes, metadata can and should be made available openly, preferably using CC0. At AUSSDA, all metadata are marked with CC0 by default.

Q&A from H2020, Horizon Europe and FWF templates

- **Will your data be made freely available in the public domain to permit the widest re-use possible? Will your data be licensed using standard reuse licenses?**
  - No, social science data itself often cannot be made available in the public domain (i.e. CC0). A Scientific Use licence is appropriate and will allow reuse for scientific purposes. Since there don’t exist universally applicable SU licences, this is a tailored licence.

Suggested wording for access conditions and licences in DMP

"Depending on the sensitivity and content of the data, we will choose licence and access conditions according to the AUSSDA standards to protect the privacy of our research subjects with the aim of reaching the highest reuse potential at the same time. Since we will collect (sensitive) personal data, access restrictions likely will be necessary to comply with data protection regulations. We will publish the data at AUSSDA according to the principle: "as open as possible, as closed as necessary".

Suggestions on data protection challenges

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How to deal with data protection challenges in RDM

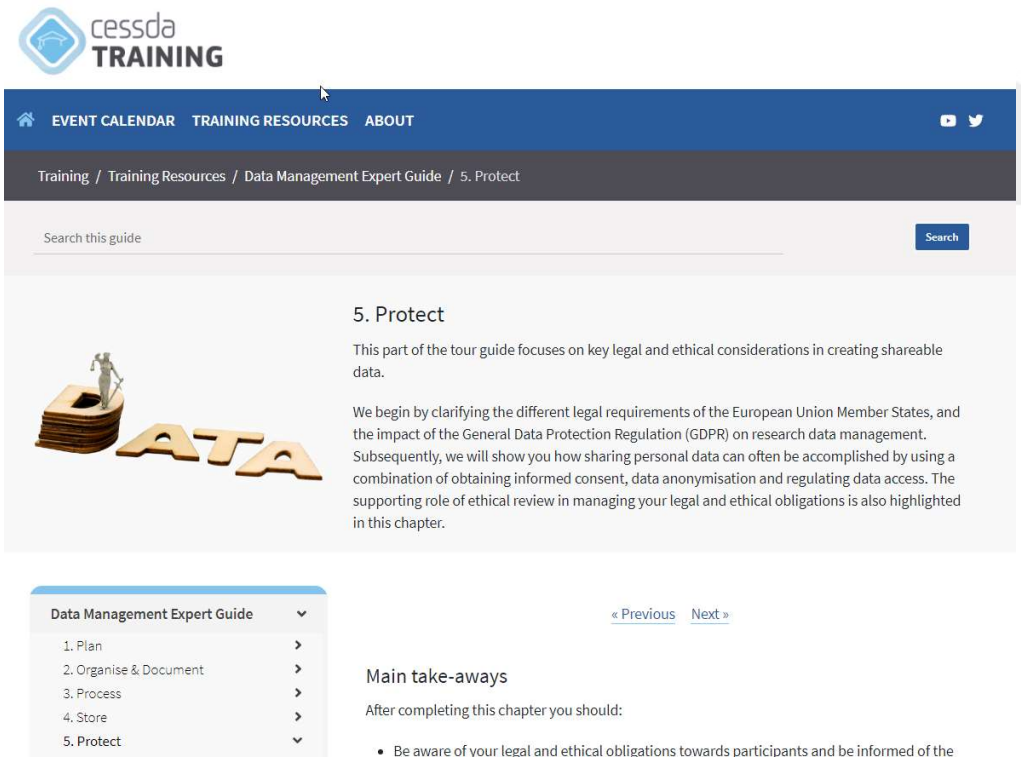
- Plan all stages of data management ahead
  - some things cannot be sought retrospectively, i.e. consent by participants or review by ethics committee
- Know applicable guidelines on data management
  - from funders, institutions, publishers etc. to avoid problems, i.e. not complying with Open Data or similar regulations
- Get in touch with infrastructures and service providers as early as possible to get information
  - i.e. check whether or not and how your data can be deposited before promising this in the DMP or ask how (not) to formulate a consent form when planning to publish your data

Some more practical suggestions

- participate in regular data protection trainings
  - usually available at universities
    - for UNIVIE - trainings at ZID and from human resources development
- consult with the data protection officer from your institution
  - for UNIVIE - ticket system for DPO for general questions on data protection
- in case of (possible) data breaches
  - consult with the DPO or the responsible office as soon as possible (as there are deadlines to be met)
  - get ahead of (some) problems: encrypt your devices and use secure technology for data transfers (i.e. Aconet file sender instead of sending data via email)

A resource: Data Management Expert Guide (DMEG)

- Additional information on how to protect your data in the social sciences
- Website: https://www.cessda.eu/Training/Training-Resources/Library/Data-Management-Expert-Guide/5.-Protect



Coffee break

cessda.eu

@CESSDA_Data

Licence: CC-BY 4.0

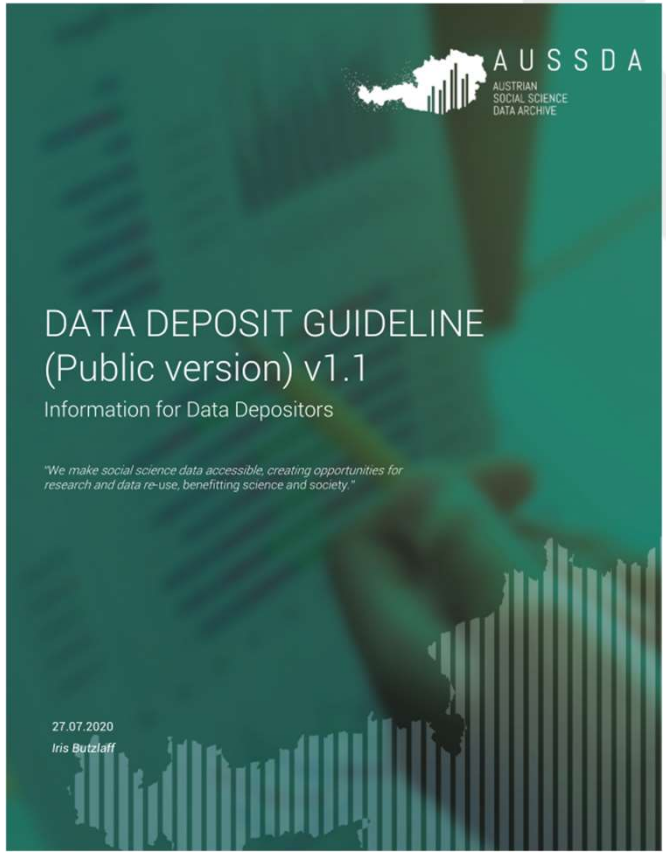
AUSSDA deposit guidelines

cessda.eu

@CESSDA_Data

Butzlaff, Iris (2020). Data Deposit Guideline (Public version) v1.1. Vienna: The Austrian Social Science Data Archive.

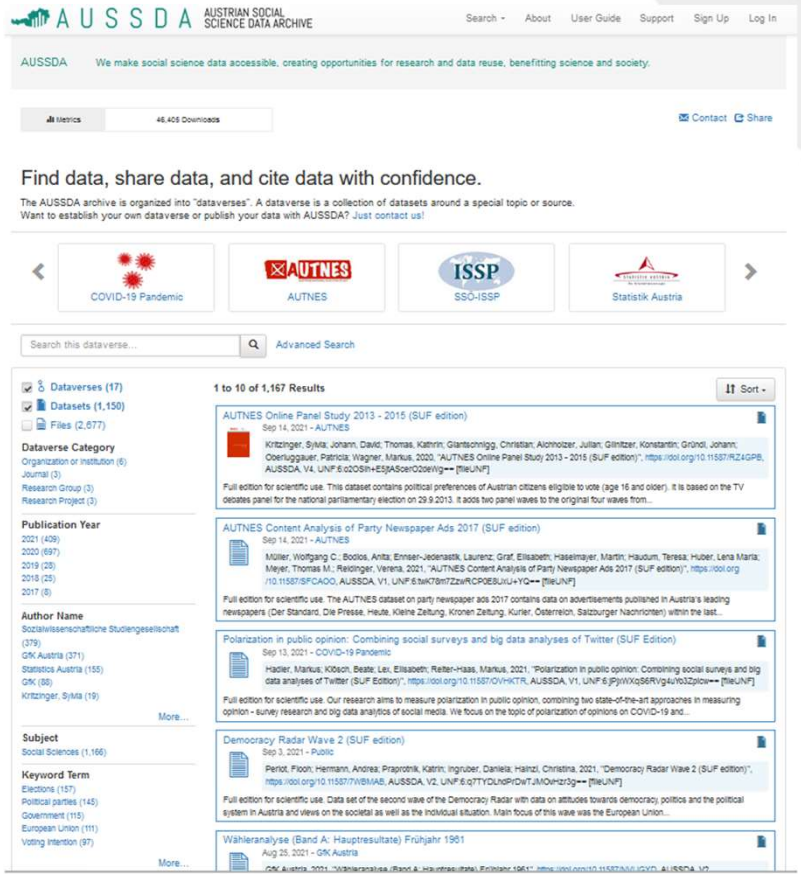
https://aussda.at/fileadmin/user_upload/p_aussda/Documents/AUSSDA-Data-Deposit-Guideline.pdf



All information on the next slides has it origins in the Data Deposit Guideline and further data handling conventions at AUSSDA.

The AUSSDA Dataverse

- https://data.aussda.at/
- Uses the open source software of the Dataverse project: https://dataverse.org/



Steps before the publication of your data

1. Get in contact and clearify the legal framework
2. Sign the contract/ get the signature for the contract
3. Prepare your data for submission
4. Submit data
5. Data curation and feedback by AUSSDA
6. Preservation
7. Publication
8. Access

Get in contact & legal framework

- Reach out to info@aussda.at regarding your publication plans
- Idealy as early as possible
- Let us know asap if you have any special conditions (deadlines, embargos…)
- Set up the contracts

Contract

- Rights holder needs to sign the contract (the rights holder could be the head of your institution, department, research project, or the data collector)
- Contract is a precondition for submitting the data

Prepare your data for publication

- Documentation material
- Data

Preparation of documentation material

Documentation material

- The more documentation material the higher the reuseability
- Usually CC-BY license
- Descriptions of the documentation material can already be part of your DMP

Documentation: mandatory

- Instruments of data collection, e.g.:
  - Questionnaire
  - Data collection guidelines….
- Metadata sheet

Documentation: recommended

- Codebook
- Methods report

Documentation: optional

- Project report
- Data Management Plan (DMP) of project proposal
- Interviewer guidelines
- Interview cards
- Documentation about incentives, contacts…
- Recoding protocol
- Informed consent forms
- Any further document that helps users to understand the data

Documentation: formats

- Preferred:
  - PDF/A
  - Plain text

- Accepted:
  - OpenDocument Text (.odt), MS Word (.docx, .doc)
  - PDF (.pdf)
  - Rich Text Format (.rtf)
  - Markdown (.md)
  - Hypertext Markup Language (.htm, .html)
  - Extensible Markup Language (.xml)
  - JavaScript Object Notation (.json)

Preparation of data

Data: preferred formats

- Stata 14 (.dta)
- SPSS (.sav)
- Tab-, or comma-delimited text files (e.g., .csv, .tab, .tsv, etc.) with command file (setup/syntax for import into Stata or SPSS)

Data: accepted formats

- OpenDocument table format (.ods), MS Excel (.xlsx, .xls)
- Tab-, or comma-delimited text files (e.g., .csv, .tab, .tsv) without command file
- Statistical software R (.R)

Check consistency and quality

- Dataset and documentation are coherent
  - Check variable names
  - Check value labels
- Check outliers for validity
- Check for plausibility

Remove all direct identifier

- social security number
- ID number from 3rd party (data collection institute)
- full name
- email address
- phone numbers
- postal codes
- date of birth
- workplace/employer
- vehicle registration number
- bank account number
- IP address
- student ID number
- passport/identity card number
- …..

Check text/string variables

- E.g. open text fields, raw text of content analysis
- Remove all direct identifiers
- Remove all personal information
- If necessary sign a form that you have completed these steps

Check demographic variables for data protection

- Keep in mind:
  - What is your universe/your population?
  - What information is essential for your data?
  - How many demographic variables are part of your data?
  - Are your data connected to other data (panel data)?
  - How sensitive is your data?
  - What are potential consequences if research subjects are re-identified?

Check demographic variables

- check for outliers
- check for observations at the fringe of continuous variables

→ recode these observations
  - E.g. age: 70 and older, householdsize: 7 and more

Check demographic variables

- Check for indirect identifiers in fine grained categories
- Check for low number of observations per category
  - Rule of thumb: recode if less than 20 observations per category

→ Recode these observations to broader categories
→ Recode these observations to a residual category („other“)

Check demographic variables: examples

- Age:
  - recode to cohorts if less than 20 observations per age group
- Education:
  - Use only ISCED on level 2
- ISCO codes:
  - 3 digit level
- NUTS codes:
  - 3 digit levels

Check demographic variables: examples

- Religious denomination
  - Recode to broader categories or „other" if less than 20 observations
- Household size, number of children:
  - recode high values to e.g. „5 and more"
- Ethnic groups, country of origin, first languages:
  - recode to groups of 20/to higher categories (e.g. higher geographic levels)

Checks in qualitative data

- Principles remain identical:
  - Remove direct identifiers
  - Check all content
  - Check for indirect identifiers

Technical requirements

- Use numerical values for missing values rather than system missings
- Do not use variable labels longer than 79 characters

Prepare your data

- Document the changes in your documentation material
- Reach out to AUSSDA if you have questions

Submit your data

- Use a secure way of sending data
  - Recommended: Aconet Filesender https://filesender.aco.net/

Curation by AUSSDA

- We check your data
- We give feedback
- We check your revised data
- We add archive variables:
  - version
  - doi

Preservation by AUSSDA

- All datafiles are converted to and published in three formats:
  - STATA (.dta)
  - SPSS (.zsav)
  - csv (.csv)

- Documentation material
  - Conversion to PDF/A
  - Machine readable variable identifier file

Q & A

What‘s next?

- Some useful STATA commands for checking your data
- Before you leave, please evaluate the workshop

Checking your data with STATA

Source: *Bodlos, Anita; Heider, Veronika; Bischof, Christian; Butzlaff, Iris (2020). Template for Data Checks. Do-File. Vienna: The Austrian Social Science Data Archive, available at: https://aussda.at/fileadmin/user_upload/p_aussda/Documents/recommended_datachecks.do.*

Before we start

- The do-file template contains commands that proved useful for data checks. This by no means indicates that all necessary checks are part of the template.
- The automated data checks are designed to help you check your data, but do not replace going through all variables and checking all variables for potential data protection violations.